



# Airoha IoT SDK for BT Data and Smart Home Get Started Guide

Version: 2.0  
Release date: 19 Mar 2026

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

## Document Revision History

| Revision | Date        | Description                                                                                                                                 |
|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0      | 29 Sep 2025 | <ul style="list-style-type: none"><li>● Initial release.</li><li>● Added get started guide for AB162x for BT Data and Smart Home.</li></ul> |
| 2.0      | 19 Mar 2026 | <ul style="list-style-type: none"><li>● Added get started guide for AB161x for BT Data – ESL.</li></ul>                                     |

## Table of Contents

|                                            |           |
|--------------------------------------------|-----------|
| <b>Document Revision History.....</b>      | <b>2</b>  |
| <b>Table of Contents.....</b>              | <b>3</b>  |
| <b>List of Figures.....</b>                | <b>3</b>  |
| <b>1 Overview .....</b>                    | <b>4</b>  |
| <b>2 Getting Started .....</b>             | <b>5</b>  |
| 2.1 Environment.....                       | 5         |
| 2.2 Developing on AB162x EVK .....         | 5         |
| 2.3 Developing on AB161x EVK .....         | 9         |
| <b>Exhibit 1 Terms and Conditions.....</b> | <b>12</b> |

## List of Figures

|                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| Figure 1. Front View of the AB162x EVK .....                                      | 6  |
| Figure 2. Downloading the Firmware to a Target Device using UART Connection ..... | 7  |
| Figure 3. Entry Dialog Box of Airoha Tool Kit .....                               | 8  |
| Figure 4. Selecting Log Binary File .....                                         | 8  |
| Figure 5. Configuring the Serial Port Parameters.....                             | 8  |
| Figure 6. Clicking the Connect Button to Open the Serial Port.....                | 8  |
| Figure 7. Clicking the Wireshark Button to View the Logs .....                    | 9  |
| Figure 8. Entry Dialog Box of Airoha Tool Kit .....                               | 10 |
| Figure 9. Selecting Log Binary File .....                                         | 10 |
| Figure 10. Configuring the Serial Port Parameters.....                            | 10 |
| Figure 11. Clicking the Connect Button to Open the Serial Port.....               | 11 |
| Figure 12. Clicking the Wireshark Button to View the Logs .....                   | 11 |

# Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide

### 1 Overview

The Airoha IoT Software Development Kit (SDK) for Bluetooth (BT) Data and Smart Home provides the software and tools for application development with the evaluation kit (EVK).

The SDK includes drivers for the hardware abstraction layer, connectivity such as Bluetooth / Bluetooth Low Energy, peripherals, and other features. It is based on the **Zephyr Operating System (OS)**.

# Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide

## 2 Getting Started

This section shows how to get started with the Airoha IoT development platform for BT Data and Smart Home. It provides information about the following items:

- Configuring the EVK
- Building the project using the SDK
- Downloading and running the project via Microsoft Windows
- Debugging the project via Microsoft Windows
- Creating your own project

Airoha IoT SDK chipsets for different product lines:

- Airoha IoT SDK for BT Data: AB162x or AB161x
- Airoha IoT SDK for Smart Home: AB162x

### 2.1 Environment

The SDK can be used with Linux (e.g., [Ubuntu 22.04 LTS](#) or a higher version), Windows OS which can support Windows Subsystem for Linux 2 (WSL 2), or macOS.

Download and extract the contents of the SDK package (e.g., IoT\_SDK\_for\_BT\_Data\_V1.0.0.ER4\_20250922) and zephyr-build-v0.26.tar.

Refer to **Airoha\_IoT\_SDK\_for\_BT\_Data\_Build\_Environment\_Guide.pdf** and **Airoha\_IoT\_SDK\_for\_Smar\_Home\_Build\_Environment\_Guide.pdf** under the `<SDK_ROOT>/mcu/doc` for the following information:

- Setup build environment
- Folder structure
- Building a project using the SDK
- Creating and configuring a project

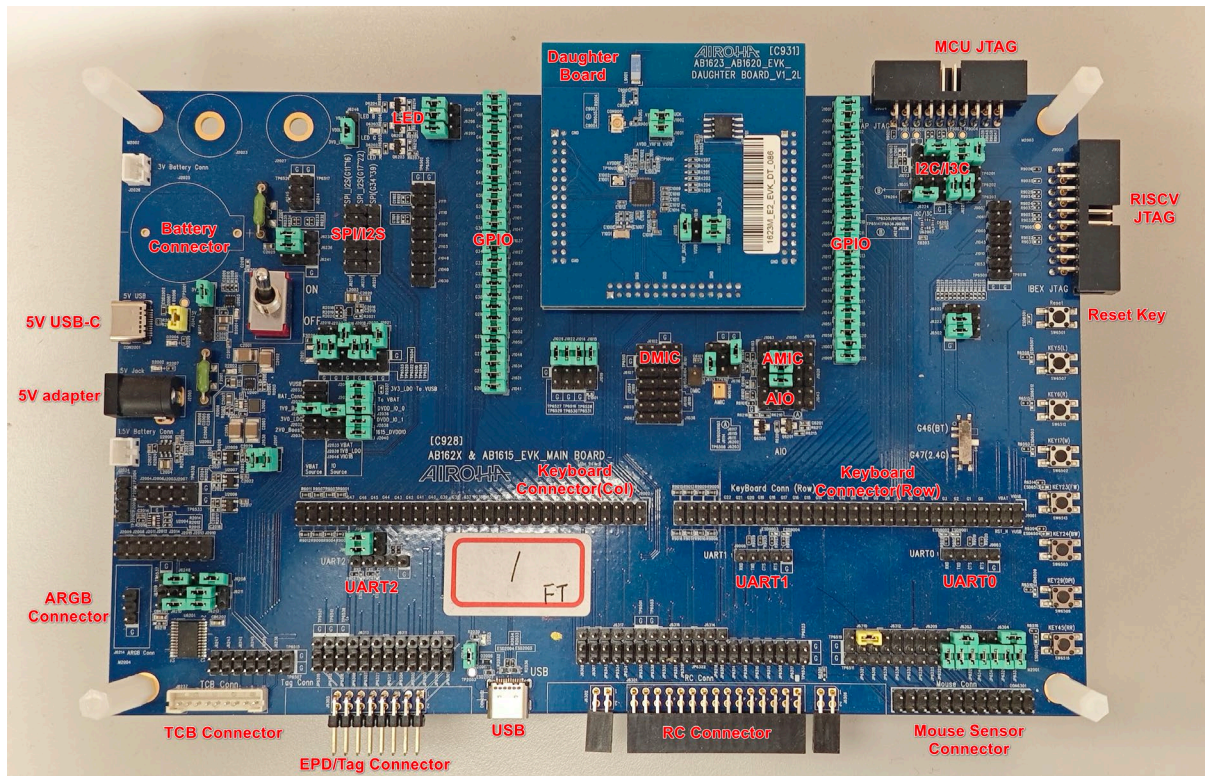
### 2.2 Developing on AB162x EVK

The following section shows the configuration and flash and run process when working with the AB162x EVK.

## Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide

## 2.2.1 Configuring the AB162x EVK



**Figure 1. Front View of the AB162x EVK**

The AB162x EVK has two separate major boards. One board is the EVK main board. The other is an adapter board for each chipset. Figure 1 shows the front view of the AB162x EVK with the AB1623 / AB1620 adapter board.

The EVK can be powered by either 5V USB-C or the 5V adapter. Set the jumpers J2002 2-3 to “on” for a 5V adapter.

AB162x has reserved two JTAG debugging interfaces. MCU\_JTAG is used for embedded CM33 and RISC-V\_JTAG is used for the embedded Ibex. Directly connect JTAG JIG with these connectors for debugging.

## 2.2.2 Flashing the Firmware on AB162x EVK

The Airoha Flash Tool is a flexible flashing tool for application development on the AB162x EVK. It is under `<SDK_ROOT>/mcu/tools/pc_tools/IOT_Flash_Tool`.

Run the `AIROHA_Flash_Tool.exe` inside this folder to start the Flash Tool.

Note that the Airoha Flash Tool is in a compressed file and needs to be unzipped before it can be used. Also, the Airoha Flash Tool relies on some basic environments. For more information, refer to [Airoha\\_Flash\\_Tool\\_on\\_Windows\\_Users\\_Guide.pdf](#).

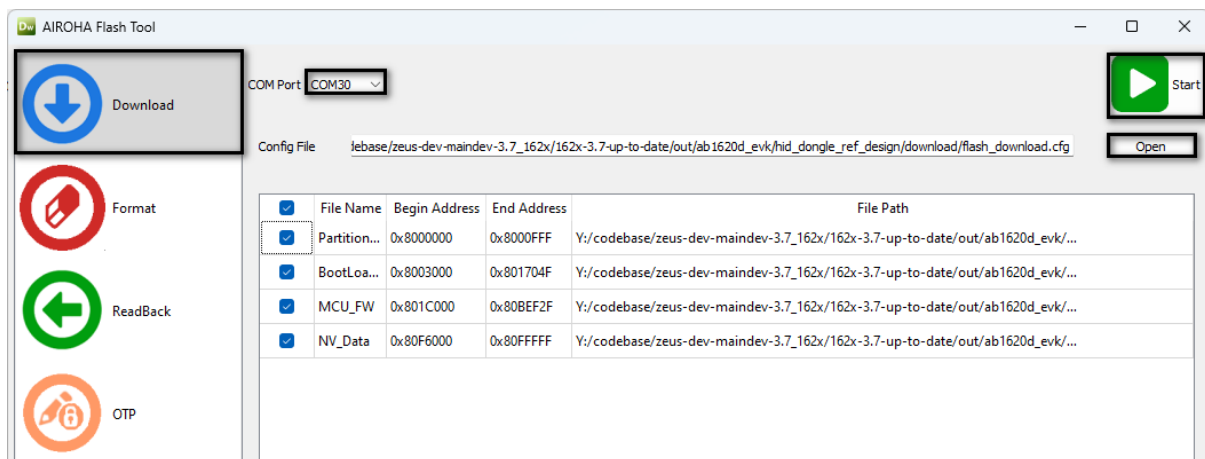
Use a pre-built download configuration file (`flash_download.cfg`) or build your own project to get one before using the Airoha Flash Tool.

To download the firmware to the target device via UART, complete the following steps:

## Airoha IoT SDK for BT Data and Smart Home

### Get Started Guide

- Connect the **UART0** cable.
- Click the **Download** button on the left panel of the main GUI.
- Select the right **UART0** from the **COM Port** drop-down menu list.
- Click the **Open** button to provide the configuration file.
- Click the **Start** button to start the download.
- Power off, then power on the target, or press the **Reset** key in the target. If the reboot is successful, the process will start automatically.



**Figure 2. Downloading the Firmware to a Target Device using UART Connection**

### 2.2.3 Running the Firmware on AB162x EVK

The Airoha Tool Kit (ATK) is a comprehensive application that allows users to view device operation logs, configure devices, conduct laboratory tests, and complete mass production (MP). It is under `<SDK_ROOT>/mcu/pc_tools/atk/ab162x`.

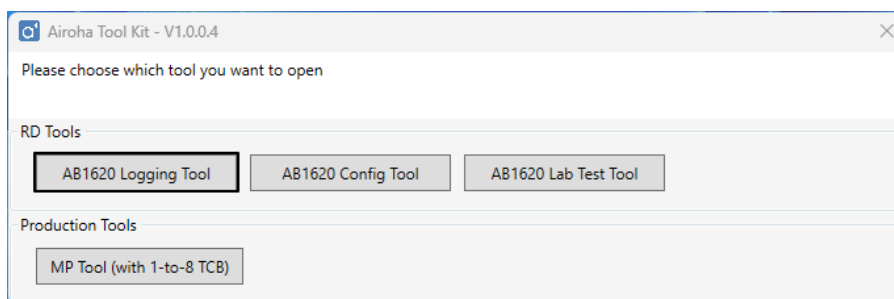
This section details how to view logs through ATK. Other functions are not described.

To use the logging tool, complete the following steps:

1. Launch the `Airoha.Tool.Kit.exe`.
2. Click the **AB1620 Logging Tool** button in the dialog box displayed by `Airoha.Tool.Kit.exe`, as shown in Figure 3.
3. Click the **Log binary file** button, as shown in Figure 4.
4. Set the Baud Rate and configure other serial port parameters, as shown in Figure 5.
5. Select the serial port from the drop-down list; click the **Connect** button, as shown in Figure 6.
6. Click the **Wireshark** button, as shown in Figure 7.
7. Reset the target. The log is displayed in the log window.

# Airoha IoT SDK for BT Data and Smart Home

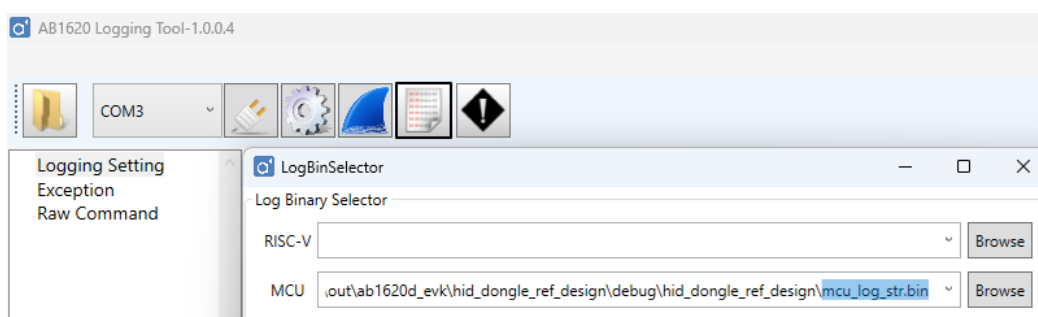
## Get Started Guide



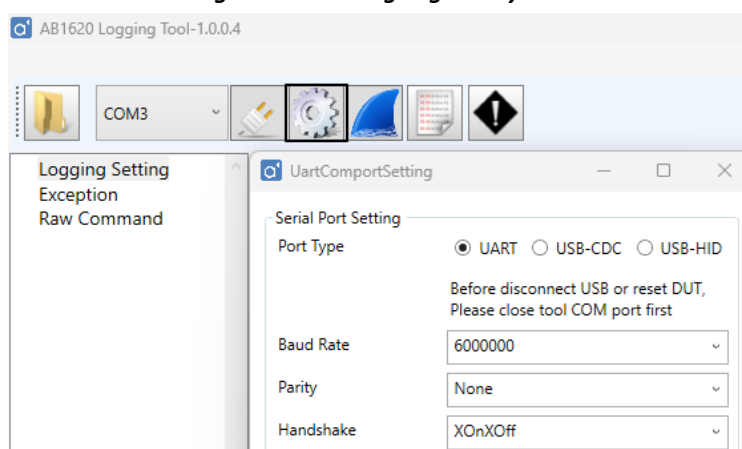
**Figure 3. Entry Dialog Box of Airoha Tool Kit**



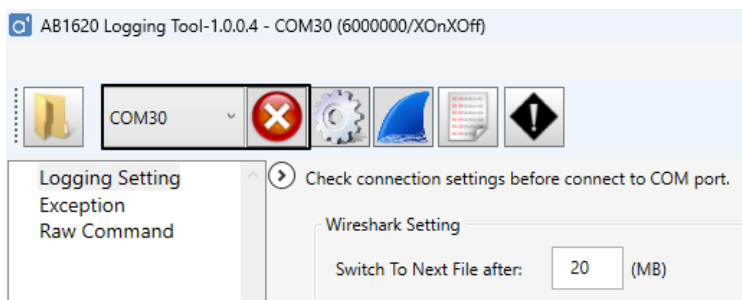
Note: The binary log file on the MCU side is **mandatory**; its filename is `mcu_log_str.bin`. The binary log file on the RISC-V side is **optional**; its filename is `risc_v_log_str.bin`.



**Figure 4. Selecting Log Binary File**



**Figure 5. Configuring the Serial Port Parameters**

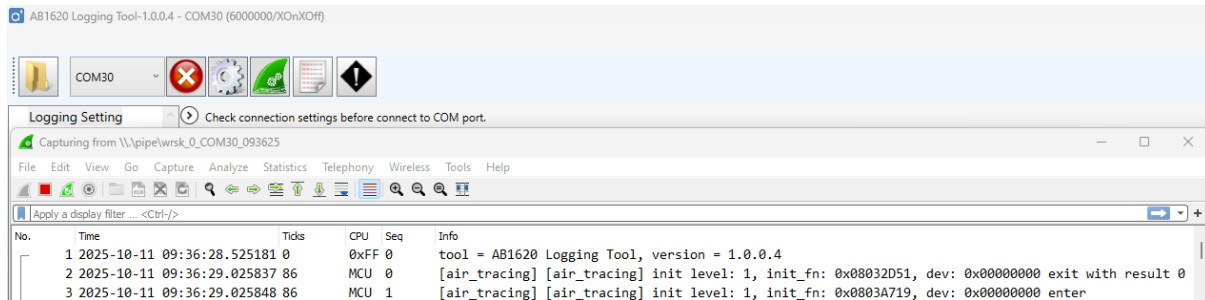


**Figure 6. Clicking the Connect Button to Open the Serial Port**



# Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide



**Figure 7. Clicking the Wireshark Button to View the Logs**

## 2.3 Developing on AB161x EVK

The following section shows the configuration and flash and run process when working with the AB161x EVK.

### 2.3.1 Configuring the AB161x EVK

Refer to the **AB1615\_AB1616\_EVK\_Users\_Guide.pdf** under `<SDK_ROOT>/mcu/doc.`

### 2.3.2 Flashing the Firmware on AB161x EVK

Refer to the **AB1615\_AB1616\_EVK\_Users\_Guide.pdf** under `<SDK_ROOT>/mcu/doc.`

### 2.3.3 Running the Firmware on AB161x EVK

The Airoha Tool Kit (ATK) is a comprehensive application that allows users to view device operation logs, configure devices, conduct laboratory tests, and complete mass production (MP). It is under `<SDK_ROOT>/mcu/pc_tools/atk/ab161x.`

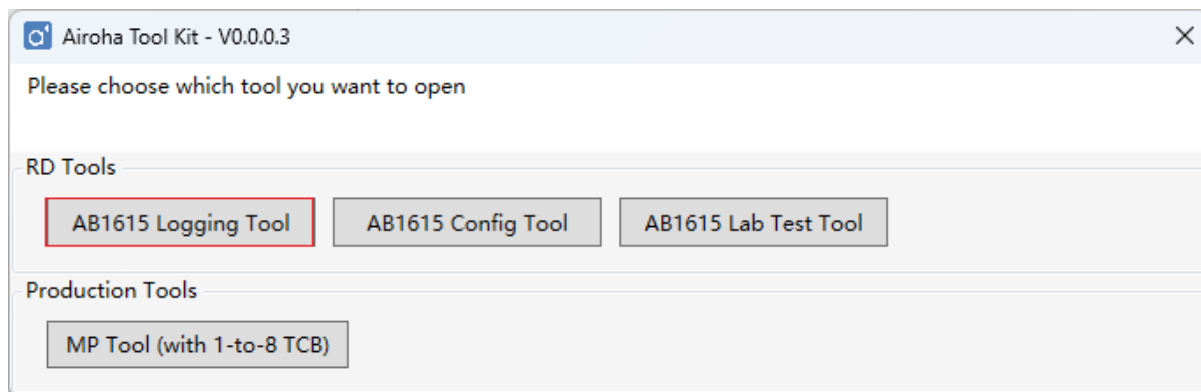
This section details how to view logs through ATK. Other functions are not described.

To use the logging tool, complete the following steps:

1. Launch the Airoha.Tool.Kit.exe.
2. Click the **AB1615 Logging Tool** button in the dialog box displayed by Airoha.Tool.Kit.exe, as shown in Figure 8.
3. Click the **Log binary file** button, as shown in Figure 9.
4. Set the Baud Rate and configure other serial port parameters, as shown in Figure 10.
5. Select the serial port from the drop-down list; then, click the **Connect** button, as shown in Figure 11.
6. Click the **Wireshark** button, as shown in Figure 12.
7. Reset the target. The log is displayed in the log window.

# Airoha IoT SDK for BT Data and Smart Home

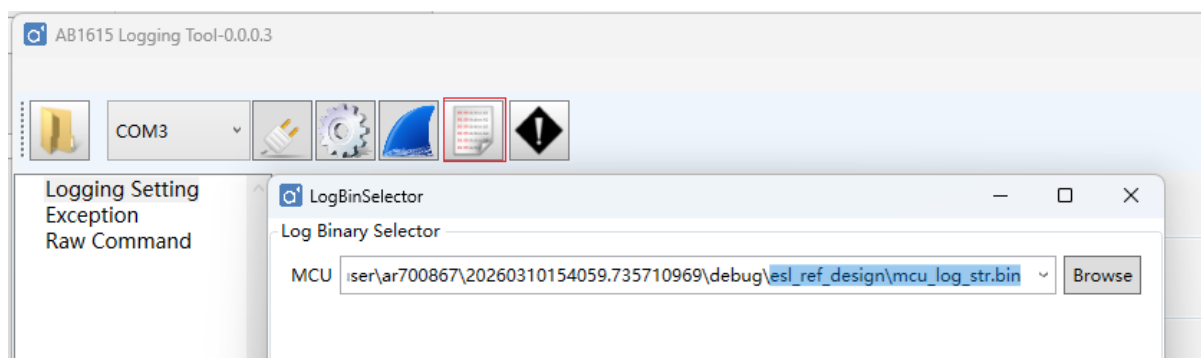
## Get Started Guide



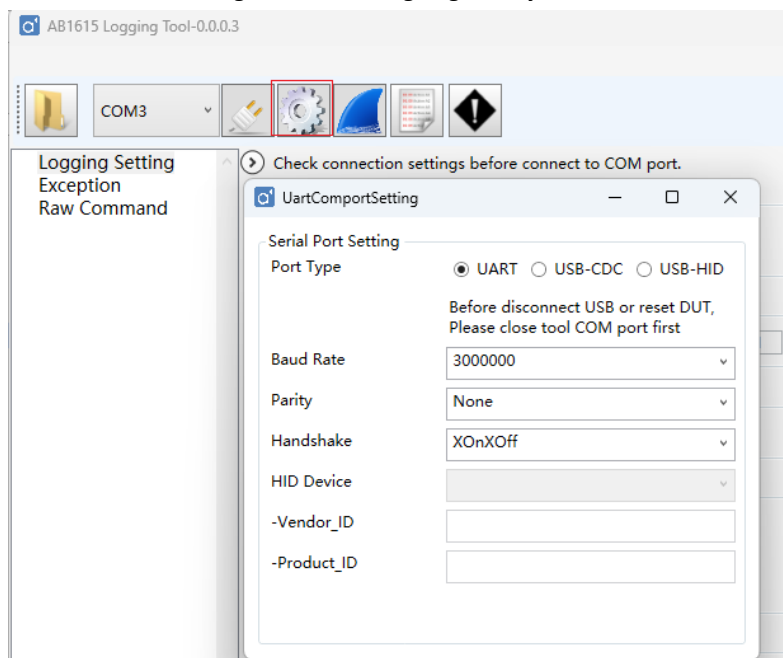
**Figure 8. Entry Dialog Box of Airoha Tool Kit**



Note: The binary log file on the MCU side is **mandatory**; its filename is `mcu_log_str.bin`.



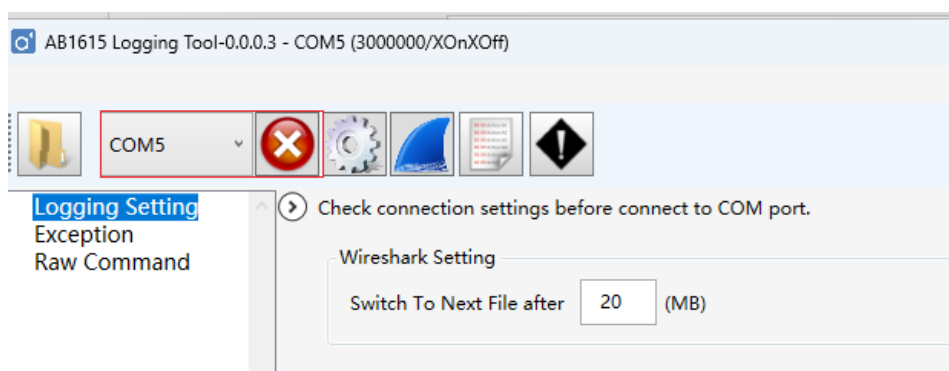
**Figure 9. Selecting Log Binary File**



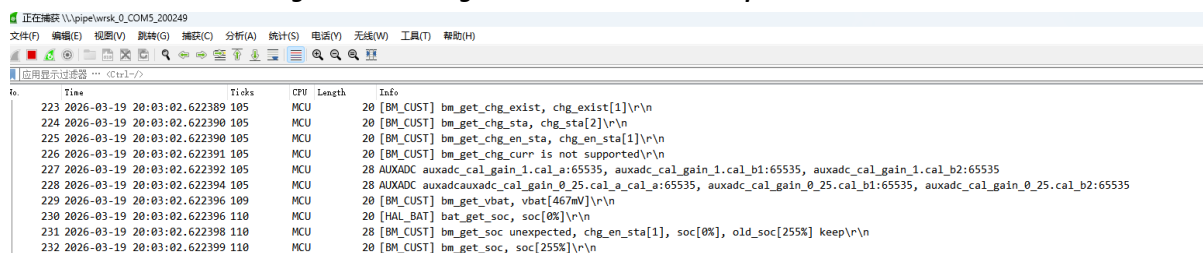
**Figure 10. Configuring the Serial Port Parameters**

# Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide



**Figure 11. Clicking the Connect Button to Open the Serial Port**



**Figure 12. Clicking the Wireshark Button to View the Logs**

# Airoha IoT SDK for BT Data and Smart Home

## Get Started Guide

### Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to Airoha Technology Corp. and/or its affiliates (collectively “Airoha”) or its licensors and is provided solely for Your internal use with Airoha’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against Airoha or any of Airoha’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify Airoha for any loss or damages suffered by Airoha for Your unauthorized use or disclosure of this Document, in whole or in part.

Airoha and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. Airoha does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY AIROHA IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. AIROHA SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. AIROHA SHALL NOT BE RESPONSIBLE FOR ANY AIROHA DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, Airoha makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does Airoha assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating Airoha’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.